When Synthetic Cells Look Too Perfect

An exploratory study of edge-based morphometrics in synthetic microscopy-like images

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INTRODUCTION

Synthetic microscopy-like images generated by modern AI systems are becoming increasingly realistic and can appear visually convincing to human observers. As such, they might have applications in biomedical sciences and healthcare [1]. However, it remains unclear whether these images preserve the structural irregularity and local heterogeneity characteristic of real biological microscopy, or whether they primarily reproduce the visual style of microscopy itself.

This exploratory Research Notes pilot compares real phase-contrast microscopy images of Caco-2 epithelial cultures [2] with synthetic microscopy-like images generated using multiple image generation systems. Rather than attempting production-grade segmentation or forensic classification, the study focuses on whether simple edge-enhanced morphology analysis can reveal differences in structural organization between real and synthetic imagery. This exploratory pilot, conducted in March 2026, examined 15 images, including two real Caco-2 microscopy references and 13 synthetic examples generated using different AI image systems and prompts.

The work evolved from an initial attempt at cell segmentation toward a broader question of structural realism. During analysis, real microscopy images frequently proved more difficult to computationally separate into clean local structures than synthetic images. This limitation itself became informative.

In particular, many synthetic microscopy-like images appeared visually plausible while simultaneously exhibiting:

- cleaner local motifs
- stronger continuous edge structure
- more repetitive geometric organization
- greater computational regularity

In contrast, the real microscopy examples often appeared biologically messy, optically imperfect, and structurally heterogeneous.

The goal of this note is not to validate a detection model, but to explore whether simple edge-based structural extraction methods can partially capture this domain gap between real and synthetic microscopy imagery.

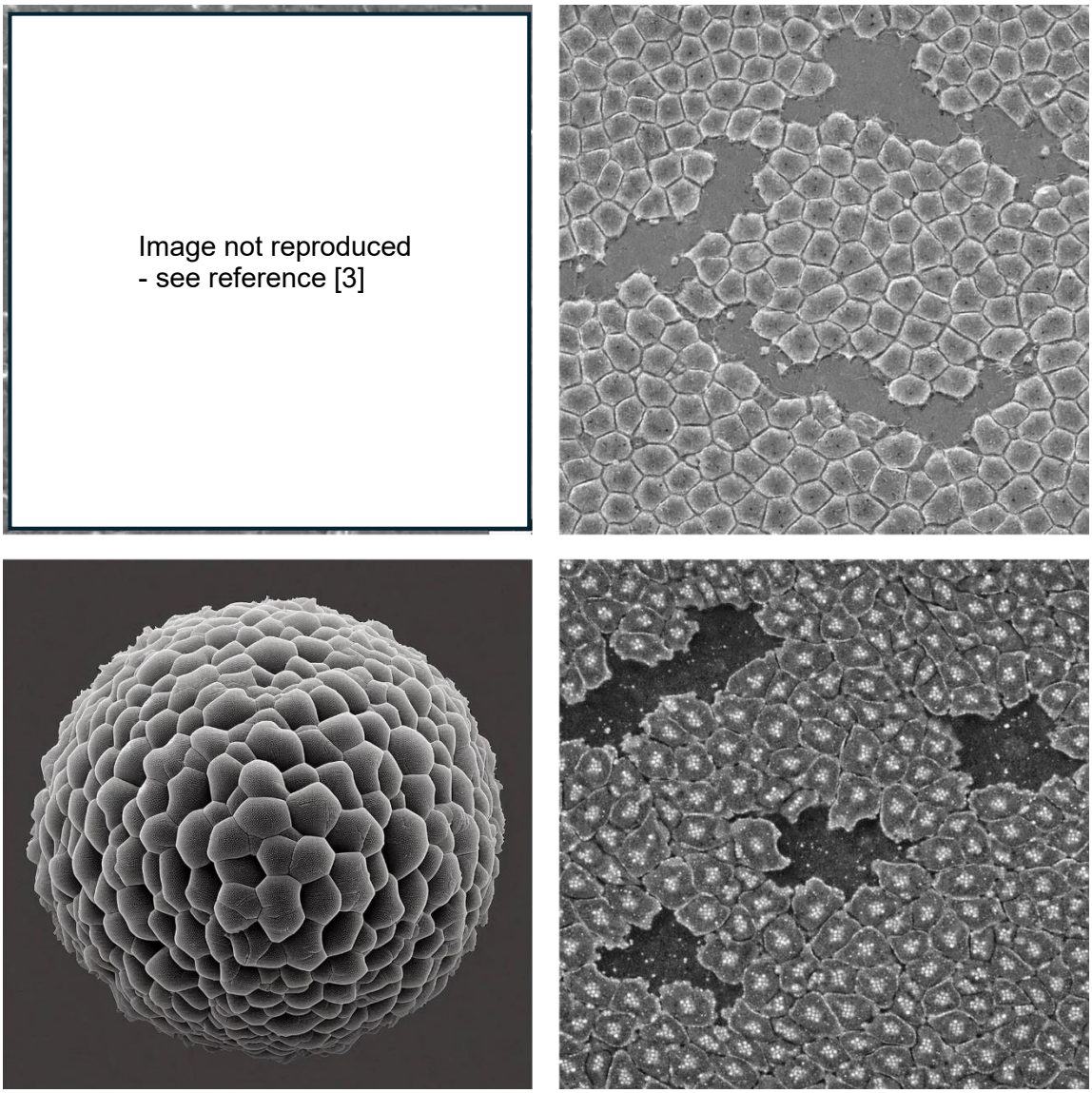


***Figure 1. Real vs synthetic microscopy-like image examples. Top left**: Real Caco-2 microscopy reference image was obtained from Cytion [3] and redacted here due to copyright. Readers may consult the original images through the cited source. **Top right:** Synthetic microscopy-like image generated using ChatGPT GPT-image. The example appears visually convincing while exhibiting unusually regular epithelial organization. **Bottom left:** Synthetic microscopy-like image generated using Bing DALL·E 3. The image contains tessellation-like or rendered spherical structures inconsistent with typical adherent epithelial morphology. **Bottom right:** Synthetic microscopy-like image generated using ChatGPT GPT-image. Despite reproducing clutter and texture, repetitive local structural patterns remain visible.*

STRUCTURAL REALISM AND EXPLORATORY WORKFLOW

Phase-contrast microscopy is widely used for observing unstained living cells and often contains optical halos, incomplete boundaries, heterogeneous texture, and local intensity variation [4]. In biological microscopy, these irregularities are not necessarily artifacts to be removed; they may reflect genuine biological and optical complexity.

Synthetic microscopy-like imagery introduces an important question: can generative systems reproduce not only the visual appearance of microscopy, but also the irregular structural organization of real biological systems?

To explore this question, a lightweight image-processing workflow was applied to both real and synthetic image groups. The analysis intentionally used simple and interpretable methods [5] rather than advanced segmentation pipelines or deep learning models.

The exploratory workflow consisted of:

- image preprocessing and grayscale standardization
- edge-enhanced structural extraction using Sobel filtering
- threshold-based edge isolation
- extraction of simple edge-derived morphometric descriptors
- qualitative visual inspection of structural organization

The goal was not precise cell segmentation. Instead, the analysis focused on whether structural extraction itself behaved differently across real and synthetic imagery.

Interestingly, the real microscopy images frequently produced fragmented or partially connected edge structures, while many synthetic images resolved into cleaner polygonal or tiled patterns under edge enhancement.

This observation became one of the central findings of the pilot study.

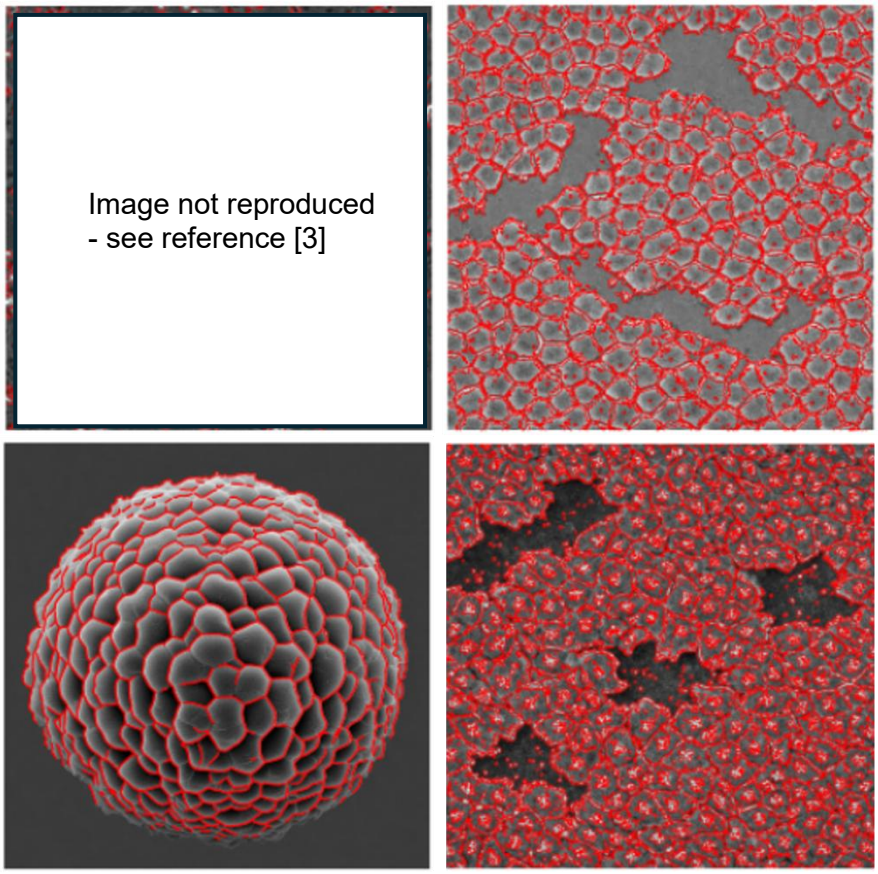


Figure 2. Edge-enhanced structural extraction of real and synthetic microscopy-like images. Note: Real Caco-2 microscopy reference image (Top left) was obtained from Cytion [3] and redacted here due to copyright. Readers may consult the original image through the cited source.

Edge-overlay visualization after Sobel-based structural extraction. Real microscopy examples frequently produced fragmented, heterogeneous edge organization, whereas several synthetic

images resolved into cleaner and more continuous local structures. The analysis was exploratory and not intended as validated cell segmentation.

QUANTITATIVE OBSERVATIONS AND DOMAIN GAP

Simple edge-derived morphometric descriptors were extracted from the image set, including:

- edge fraction
- number of connected edge objects
- object area
- perimeter
- eccentricity
- solidity

Scatter plot analysis suggested some differences between the real and synthetic examples, although considerable overlap remained.

Importantly, the observed domain gap was not merely related to pixel intensity variance or image noise. Many synthetic images retained substantial visual complexity while still appearing:

- geometrically regular
- locally repetitive
- computationally organized
- easier to structurally separate into repeated motifs

In contrast, the real microscopy examples often exhibited:

- fragmented boundaries
- heterogeneous local texture
- optical irregularity
- incomplete structural continuity

These characteristics complicated edge-based extraction while simultaneously contributing to the biological realism of the images.

Qualitative visual inspection revealed differences in spatial organization and texture that were not fully represented by the simple measurements used in this pilot.

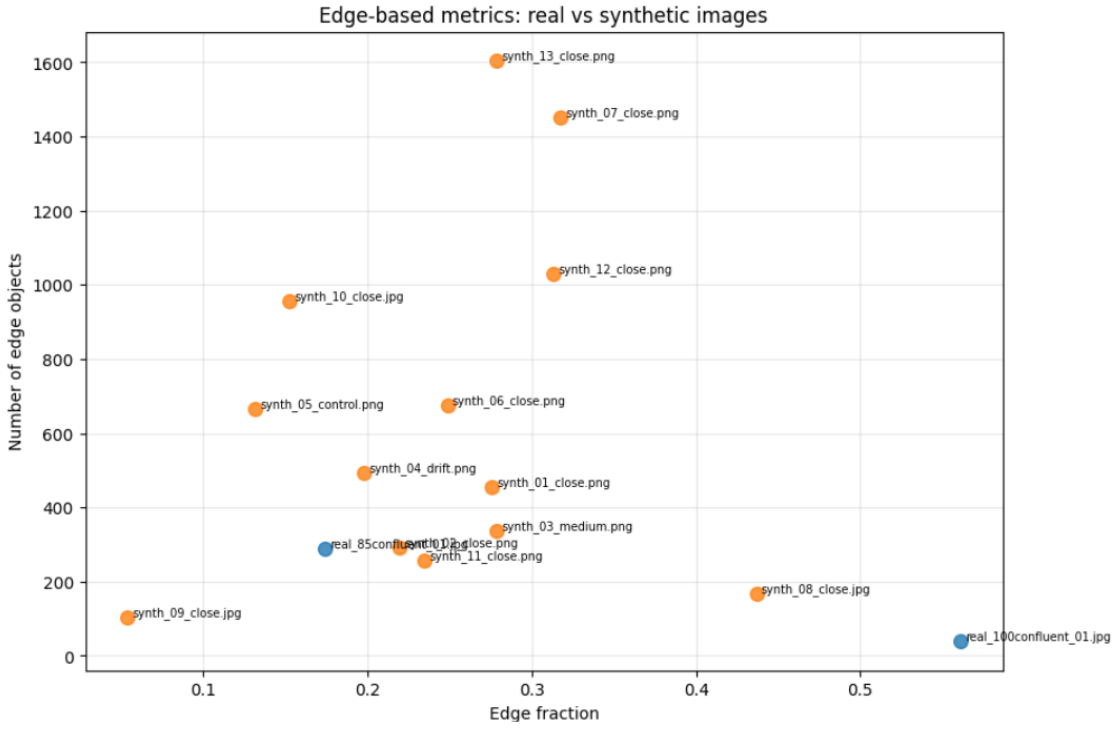


Figure 3. Edge-derived morphometric comparison. Scatter plot of edge fraction versus number of connected edge objects across real and synthetic microscopy-like images.

Partial separation between image groups was observed, although substantial overlap remained. The results suggest that simple edge-derived descriptors capture part, but not all, of the structural differences apparent through qualitative visual inspection.

INTERPRETATION, LIMITATIONS, AND CONCLUSION

This exploratory pilot suggests that synthetic microscopy-like images may reproduce aspects of visual appearance while simplifying the structural irregularity and optical heterogeneity present in real biological microscopy.

Interestingly, real microscopy images may be difficult precisely because biological systems are heterogeneous and optically imperfect. Synthetic images, in contrast, may unintentionally simplify this complexity into cleaner computational structure while still appearing visually convincing. Several synthetic examples were generated using prompts explicitly attempting to reproduce epithelial morphology and phase-contrast microscopy appearance. Yet despite achieving visual plausibility, the resulting structures often remained more computationally regular than the real microscopy reference images. This suggests that biological realism may involve organizational characteristics that are difficult to fully capture through textual description alone.

The study also highlights an important distinction between:

- visual plausibility and
- higher-order structural realism.

Although simple edge-based morphometric descriptors partially captured the domain gap between real and synthetic imagery, qualitative visual interpretation remained important. Qualitative visual

inspection revealed differences in spatial organization and texture that were not fully represented by the low-dimensional features extracted in this pilot.

This work has several limitations:

- small exploratory dataset
- limited number of real microscopy examples
- no ground-truth segmentation
- lightweight image-processing methods only
- multiple synthetic generation styles

The note is therefore intended as an exploratory Research Notes study rather than a benchmark computer vision pipeline or forensic AI detection framework.

Future work could explore larger microscopy datasets, more controlled comparisons across image generators, and methods capable of capturing more complex spatial patterns.

Despite its limitations, this exploratory pilot suggests that simple edge-enhanced morphology analysis may help reveal structural differences between real and synthetic microscopy-like imagery.

More broadly, the observations suggest that biological messiness itself may contain important structural information that synthetic imagery does not yet fully reproduce.

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